Cyclic Groups

Semester Assignment

1 General Integral Powers

Before diving into cyclic groups, we must understand how we define the "power" of an element within a group. Let $(G,*)$ be a group and $a \in G$. For any positive integer n, we define the n-th power of a as the element resulting from operating a with itself n times.

Definitions

1. **Positive Powers:** For $n \in \mathbb{Z}^+$, $a^n = a * a * \cdots * a$ (n times).
2. **Zero Power:** $a^0 = e$, where e is the identity element of the group G.
3. **Negative Powers:** For $n \in \mathbb{Z}^+$, $a^{-n} = (a^n)^{-1} = (a^{-1})^n$.

Examples

- **Example 1 (Multiplicative Group):** In the group $(\mathbb{R} \setminus \{0\}, \cdot)$, if $a = 2$ and $n = 3$, then $2^3 = 2 \cdot 2 \cdot 2 = 8$.
- **Example 2 (Additive Group):** In an additive group $(\mathbb{Z},+)$, "power" is expressed as $n \cdot a$. If $a = 5$ and $n = 3$, then $3(5) = 5+5+5 = 15$.

2 Cyclic Groups

A group G is called a **cyclic group** if every element in the group can be expressed as an integral power of a single element $a \in G$. In this case, the element a is known as the **generator** of the group.

We denote a cyclic group generated by a as $G = \langle a \rangle$. Mathematically:

$$G = \{a^n : n \in \mathbb{Z}\}$$

Finite and Infinite Cyclic Groups

Depending on the number of elements the generator produces, cyclic groups are classified into two types:

1. **Infinite Cyclic Group:** If all integral powers of the generator a (i.e., $\dots, a^{-2}, a^{-1}, a^0, a^1, a^2, \dots$) result in distinct elements, the group is an infinite cyclic group.

2. **Finite Cyclic Group:** If there exists a positive integer n such that $a^n = e$, the powers will eventually start repeating. The set of elements will be finite, and the group is called a finite cyclic group.

Examples

- **Example 1 (Infinite):** The group $(\mathbb{Z}, +)$ is an infinite cyclic group. It is generated by the element 1 because every integer can be written as $n \cdot 1$. (Note: -1 is also a generator).
- **Example 2 (Finite):** The group $(\mathbb{Z}_4, +_4) = \{0, 1, 2, 3\}$ is a finite cyclic group of order 4. It is generated by 1 since: $1^1 = 1, 1^2 = 2, 1^3 = 3, 1^4 = 0$.

3 Number of Generators of a Cyclic Group

The number of generators in a cyclic group depends on whether the group is infinite or finite.

Definition: Euler's Totient Function $\phi(n)$

To find the number of generators for a finite group, we use **Euler's Function**, $\phi(n)$. This function counts the number of positive integers less than n that are relatively prime to n (i.e., $gcd(k, n) = 1$).

1. **For Infinite Cyclic Groups:** An infinite cyclic group has exactly **two** generators. If a is a generator, then a^{-1} is the only other generator.
2. **For Finite Cyclic Groups:** If G is a finite cyclic group of order n, then the number of generators is given by $\phi(n)$. Any element a^k is a generator if and only if $gcd(k, n) = 1$.

Examples

- **Example 1:** Consider a cyclic group G of order $n = 6$. The integers less than 6 and prime to 6 are $\{1, 5\}$. Thus, $\phi(6) = 2$. If a is a generator, the generators are a^1 and a^5.
- **Example 2:** Consider a cyclic group G of order $n = 7$. Since 7 is a prime number, all integers $\{1, 2, 3, 4, 5, 6\}$ are relatively prime to 7. Thus, $\phi(7) = 6$, meaning the group has 6 different generators.

4.3 Properties of Cyclic Groups

Property 1: Inverse as a Generator

Theorem: If $G = \langle a \rangle$ is a cyclic group, then $G = \langle a^{-1} \rangle$.

Let G be a cyclic group generated by a. Let $x \in G$ be an arbitrary element. By definition, $x = a^n$ for some integer $n \in \mathbb{Z}$. We can write:

$$x = (a^{-1})^{-n}$$

Since n is an integer, $-n$ is also an integer. This shows that any element x in G can be expressed as an integral power of a^{-1}. Therefore, a^{-1} is also a generator of G. Thus, $G = \langle a^{-1} \rangle$.

Question: Identify the generators of the additive group of integers $(\mathbb{Z}, +)$.

Solution: The group $(\mathbb{Z}, +)$ is generated by 1, as any integer k can be written as $k(1)$. By Property 1, the inverse of 1 must also be a generator. In $(\mathbb{Z}, +)$, the inverse of 1 is -1. Thus, the generators are $\{1, -1\}$.

Property 2: Abelian Nature

Theorem: Every cyclic group is an Abelian group.

Let $G = \langle a \rangle$ be a cyclic group. Let x, y be any two elements in G. Then $x = a^r$ and $y = a^s$ for some integers $r, s \in \mathbb{Z}$. Consider the product:

$$x * y = a^r * a^s = a^{r+s}$$

Since addition of integers is commutative ($r + s = s + r$), we have:

$$a^{r+s} = a^{s+r} = a^s * a^r = y * x$$

Since $x * y = y * x$, the group is Abelian.

Question: Is every Abelian group cyclic?

Solution: No. Consider the Klein 4-group $V = \{e, a, b, c\}$. It is Abelian because its operation table is symmetric, but it is not cyclic because no single element can generate all four elements.

Property 3: Order of Group and Generator

Theorem: The order of a finite cyclic group is equal to the order of its generator.

Let $G = \langle a \rangle$ and let $O(a) = n$. This means n is the smallest positive integer such that $a^n = e$. The elements $a^1, a^2, \ldots, a^n = e$ are all distinct. If $a^i = a^j$ for $1 \leq j < i \leq n$, then $a^{i-j} = e$, where $i - j < n$, contradicting that n is the smallest such integer. Every power a^m can be reduced to a^r where $0 \leq r < n$ using the division algorithm. Thus, G has exactly n elements. Hence, $O(G) = n = O(a)$.

Question: Find the order of the group generated by i in the complex numbers under multiplication.

Solution: $i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1$. The order of the generator i is 4. By Property 3, the group $G = \{i, -1, -i, 1\}$ has order 4.

Property 4: Subgroups of Cyclic Groups

Theorem: Every subgroup of a cyclic group is cyclic.

Let $G = \langle a \rangle$ and H be a subgroup of G. If $H = \{e\}$, it is cyclic. If not, let m be the smallest positive integer such that $a^m \in H$. For any $a^k \in H$, write $k = mq + r$ with $0 \le r < m$. Then $a^r = a^{k-mq} = a^k * (a^m)^{-q}$. Since $a^k, a^m \in H$, then $a^r \in H$. By the minimality of m, r must be 0. Thus $k = mq$ and $H = \langle a^m \rangle$.

Question: List the subgroups of $(\mathbb{Z}_6, +_6)$.

Solution: Divisors of 6 are 1, 2, 3, 6. Subgroups are: $\langle 0 \rangle = \{0\}$, $\langle 3 \rangle = \{0,3\}$, $\langle 2 \rangle = \{0,2,4\}$, and $\langle 1 \rangle = \mathbb{Z}_6$. All are cyclic.

Property 5: Order of a Finite Cyclic Group

Theorem: The order of a finite cyclic group is equal to the order of its generator. i.e., If $G = \langle a \rangle$, then $O(G) = O(a)$.

Let $G = \langle a \rangle$ be a cyclic group and let the order of the generator a be n. By definition of the order of an element, n is the least positive integer such that $a^n = e$ (where e is the identity element).

We will first show that the set $S = \{a^1, a^2, a^3, \ldots, a^n = e\}$ contains n distinct elements. Suppose, if possible, that any two elements in S are equal. Let $a^i = a^j$ where $1 \le j < i \le n$. Multiplying both sides by a^{-j}, we get:

$$a^i \cdot a^{-j} = a^j \cdot a^{-j} \implies a^{i-j} = e$$

However, since $1 \le j < i \le n$, it follows that $0 < i - j < n$. This contradicts our assumption that n is the *least* positive integer such that $a^n = e$. Therefore, all n elements in S must be distinct.

Next, we show that any arbitrary element $a^m \in G$ is equal to one of the elements in S. By the division algorithm, for any integer m, there exist integers q and r such that:

$$m = nq + r, \quad \text{where } 0 \le r < n$$

Then,

$$a^m = a^{nq+r} = (a^n)^q \cdot a^r = e^q \cdot a^r = a^r$$

If $r = 0$, then $a^r = a^n = e \in S$. If $0 < r < n$, then $a^r \in S$. In either case, every element of G is contained in S. Since S contains exactly n distinct elements, the order of the group G is n. Hence, $O(G) = n = O(a)$.

Question: Consider the group $G = \{1, -1, i, -i\}$ under the operation of multiplication. Verify that the order of the group is equal to the order of its generator.

Solution: The group G has 4 elements, so $O(G) = 4$. The group is cyclic and generated by i. Let us find the order of i:

- $i^1 = i$
- $i^2 = -1$

- $i^3 = -i$
- $i^4 = 1$ (Identity element)

The smallest positive integer n such that $i^n = 1$ is 4. Therefore, $O(i) = 4$. Since $O(G) = 4$ and $O(i) = 4$, we have $O(G) = O(i)$, which verifies the theorem.

4.4 Detailed Corollaries and Applications

Corollary 1: Subgroups of Infinite Cyclic Groups

Statement: Every proper subgroup of an infinite cyclic group is also an infinite cyclic group.

Let $G = \langle a \rangle$ be an infinite cyclic group and H be a proper subgroup of G ($H \neq \{e\}$). From the theorem that "Every subgroup of a cyclic group is cyclic," we know $H = \langle a^m \rangle$ for some smallest positive integer m. Since G is infinite, $a^m \neq e$ for any $m > 0$. If H were finite, there would exist j, k such that $(a^m)^j = (a^m)^k$, implying $a^{m(j-k)} = e$, which contradicts the infinite nature of G. Thus, H must be infinite.

Converse: The converse states: "If every proper subgroup of a group is infinite cyclic, then the group itself is infinite cyclic." This is **not necessarily true** in general group theory, but for a group to have only infinite cyclic subgroups, the group itself must be infinite.

Question: Let $G = (\mathbb{Z}, +)$ and $H = 5\mathbb{Z}$. Prove H is an infinite cyclic subgroup and find its generators.

Solution: G is an infinite cyclic group generated by 1. $H = \{\dots, -10, -5, 0, 5, 10, \dots\}$ is the set of all multiples of 5. 1. Since H consists of all elements $(1+1+1+1+1)^n$, $H = \langle 5 \rangle$. 2. As there are no two distinct integers j, k such that $5j = 5k$, the group is infinite. By Corollary 1, H is infinite cyclic. Its generators are 5 and -5.

Corollary 2: Existence of an Element of Order n

Statement: A finite group G of order n is cyclic if and only if there exists an element $a \in G$ such that $O(a) = n$.

($\implies$) If G is cyclic of order n, by definition $G = \langle a \rangle$. The set $\{a, a^2, \dots, a^n = e\}$ must contain n distinct elements, so the smallest power resulting in e is n. Thus $O(a) = n$.
($\impliedby$) If there exists $a \in G$ such that $O(a) = n$, then the cyclic subgroup $\langle a \rangle = \{a, a^2, \dots, a^n\}$ has n elements. Since G only has n elements, $G = \langle a \rangle$.

Converse: The converse is inherently part of the "if and only if" statement. It implies that if a group has no element of order n, it cannot be cyclic.

Question: Show that the Klein 4-group $V = \{e, a, b, c\}$, where every non-identity element has order 2, is not cyclic.

Solution: 1. The order of the group is $O(V) = 4$. 2. For V to be cyclic, by Corollary 2, there must exist at least one element of order 4. 3. However, the problem states (and we can verify via the group table) that $O(a) = 2$, $O(b) = 2$, and $O(c) = 2$. Since no element has order 4, the group V fails the condition of Corollary 2 and is therefore **not cyclic**.